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From accurate to actionable: Interpretable $PM_{2.5}$ forecasting with feature engineering and SHAP for the Liverpool–Wirral region

Seyed Matin Malakouti

Amirkabir University of Technology, Tehran, Iran

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ABSTRACT

Fine particulate matter ($PM_{2.5}$) poses a significant threat to public health worldwide, contributing to millions of premature deaths annually, according to the World Health Organization. At the regional scale, Europe and the United Kingdom continue to experience $PM_{2.5}$ episodes influenced by transboundary pollution, industrial emissions, and meteorological patterns. Locally, the Liverpool–Wirral area faces specific challenges due to dense urban traffic, port activities, and mixed industrial–residential land use, which can lead to localized pollution hotspots.

Despite advances in air quality modeling, there remains a gap in producing highly accurate and interpretable short-term $PM_{2.5}$ forecasts tailored to local conditions using dense networks of low-cost sensors. To address this gap, machine learning models—ExtraTrees, LightGBM, and a weighted ensemble—were developed in this study to forecast daily $PM_{2.5}$ concentrations from 2019 to 2024 using a rich set of engineered time-series features. These features included lagged values (1, 7, 14, 30 days), rolling averages (3, 7, 14, 30 days), day-over-day raw and percentage changes, day of the week, weekend indicator, month, and cyclical day-of-year components (sine and cosine) to capture short-term autocorrelation, medium- and long-term trends, and seasonal effects.

The models were trained and validated on a Liverpool–Wirral dataset, and their performance was evaluated on early-2024 observations. To interpret feature contributions, SHAP (SHapley Additive exPlanations) values were computed for the LightGBM model, revealing that the 3-day rolling mean, day-over-day change, and 1-day lag dominated the predictive power. The ensemble model achieved the lowest test-set RMSE (0.54 $\mu g/m^3$, $R^2 = 0.990$). A full-year 2025 forecast indicated modest seasonal variability, with ensemble predictions remaining stable around 6–6.2 $\mu g/m^3$. These results demonstrate that careful feature engineering, coupled with SHAP-based interpretation, can yield highly accurate and transparent $PM_{2.5}$ forecasts.

Introduction

In recent years, air quality monitoring has gained significant aAir quality monitoring is increasingly vital due to its effects on health and the environment. Numerous studies have contributed to datasets and methodologies for assessing air pollution. For instance, Diez et al. (2024) provided a comprehensive multi-city air quality dataset using commercial low-cost sensors and reference data for evaluation (Diez et al., 2024). Similarly, Van Poppel et al. (2023) contributed a multi-city dataset from European cities, focusing on low-cost sensor systems (Van Poppel et al., 2023). Sserunjogi et al. (2022) expanded this approach by utilizing a spatially distributed AirQo network in Kampala, Uganda, offering neighborhood-level insights (Sserunjogi et al., 2022).

Wei et al. (2023) presented a year-long dataset collected from residential air purifiers in China, revealing crucial data for indoor air quality studies (Wei et al., 2023). Mak and Wong (2020) further developed smart city air quality data integration platforms, providing an open-source dataset for global air quality monitoring (Mak and Wong, 2020). Lee and Anderson (2022) established the OpenAQ platform, which offers a standardized global air quality repository (Lee and Anderson, 2022). Zhang et al. (2021) also contributed a dataset on air quality monitoring using IoT devices from 2018 to 2020 (Zhang et al., 2021).

Kong et al. (2021) conducted a six-year high-resolution reanalysis of air quality data across China, using surface observations from CNEMC (Kong et al., 2021). Li et al. (2024) introduced a framework for mitigating sample-imbalance in PM2.5 retrieval, providing more accurate data (Li et al., 2024). Wang et al. (2024) created a long-term daily PM2.5 dataset for India, spanning from 1980 to 2022, which is invaluable for health and environmental studies (Wang et al., 2024). Hao et al. (2024) used visibility data to derive aerosol optical depth

E-mail address: matin.malakouti@aut.ac.ir.

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(AOD) over global land, contributing to climate and satellite air quality studies (Hao et al., 2024).

The work by van Donkelaar et al. (2022) provided global estimates of PM2.5 at 1 km resolution for 1998–2019, which has been crucial for assessing air pollution impacts (van Donkelaar et al., 2022). Xiao et al. (2022) integrated satellite-derived NO2 data with ground observations, offering valuable insights into global air pollution (Xiao et al., 2022). Liu et al. (2021) demonstrated the effectiveness of spatiotemporal fusion of satellite and ground data to estimate PM2.5 over China (Liu et al., 2021). Zhang et al. (2022) focused on the impact of agricultural burning emissions in Southeast Asia, providing a high-resolution dataset for regional air quality analysis (Zhang et al., 2022).

Bhatia et al. (2023) compiled a comprehensive urban air quality dataset for India, highlighting the regional differences and challenges in air quality management (Bhatia et al., 2023). Marchenko et al. (2021) presented a long-term Arctic dust dataset, which has been invaluable for polar climate studies (Marchenko et al., 2021). McDuffie and Chang (2022) provided a synchronized dataset of wildfire emissions and ambient air quality in the western United States (McDuffie and Chang, 2022). Shang and Zhou (2022) created a 20-year high-resolution dataset of urban heat island intensity and associated air quality impacts, aiding urban environmental research (Shang and Zhou, 2022).

Fu et al. (2021) presented airborne observations of PM2.5 across the United States, offering valuable comparisons with ground-based data (Fu et al., 2021). Tang et al. (2021) contributed a global dataset of surface PM10 concentrations at $0.1° \times 0.1°$ resolution, offering a high level of spatial resolution for air quality studies (Tang et al., 2021). Liu et al. (2023) provided a high-resolution global dataset for NOx emissions from power plants, useful for air quality modeling (Liu et al., 2023). The work by Petropoulos and Akritidis (2021) combined satellite and ground observations for global land surface PM2.5 estimates, enhancing the accuracy of air quality monitoring (Petropoulos and Akritidis, 2021).

Reani et al. (2022) contributed UK daily meteorology, air quality, and pollen measurements, using statistical imputation to address missing data (Reani et al., 2022). Gledson et al. (2021) developed the Britain Breathing dataset, focusing on regional air quality estimates for the UK (Gledson et al., 2021). Lowe et al. (2021) introduced the Britain Breathing air quality dataset tools, which facilitate the analysis of air quality data across the UK (Lowe et al., 2021). Cao (2024) integrated ten years of weather, traffic, and air pollution data from Norway's largest cities, providing valuable data for urban air quality analysis (Cao, 2024).

Kumar et al. (2025) created a long-term high-resolution air quality reanalysis over the United States, providing a public-facing air quality dashboard for real-time monitoring (Kumar et al., 2025). Hoinaski et al. (2024) developed the Brazilian Atmospheric Inventories (BRAIN), a comprehensive database of air quality in Brazil, useful for regional environmental studies (Hoinaski et al., 2024). Yatkin et al. (2022) focused on a citizen-centered urban network for weather and air quality monitoring in Greenwich, offering valuable lessons for low-resource settings (Yatkin et al., 2022). Bedkari et al. (2023) provided an R package and Shiny application for visualizing and accessing global air quality data, facilitating data analysis (Bedkari et al., 2023).

De Meo et al. (2022) enhanced urban air quality monitoring in Milan through distributed low-cost sensors, offering valuable insights into urban pollution patterns (De Meo et al., 2022). Liu et al. (2021) used satellite and ground-based data fusion for PM2.5 estimation over China, providing higher accuracy for regional air quality assessments (Liu et al., 2021). Wu and Collett (2023) assessed the impact of high-resolution inputs on the North American air quality forecast capability dataset, contributing to model improvements (Wu and Collett, 2023). Zhang et al. (2022) studied the effects of agricultural burning emissions on air quality in Southeast Asia, enhancing regional policy efforts (Zhang et al., 2022).

Bhatia et al. (2023) provided a comprehensive dataset on urban air quality in India, addressing key challenges and gaps in data (Bhatia et al., 2023). Finally, Marchenko et al. (2021) focused on Arctic dust data, providing valuable insights for climate research (Marchenko et al., 2021). ttention due to its impact on public health and climate. Several studies have contributed to the development of datasets and models for understanding air pollution dynamics. For instance, the study by Diez et al. (2024) provided a comprehensive multi-city air quality dataset collected through commercial low-cost sensors, offering valuable insights for performance evaluation. Similarly, Van Van Poppel et al. (2023) collected data across multiple European cities using open-source, low-cost sensor systems, helping researchers address gaps in air quality monitoring for urban regions. Sserunjogi et al. (2022) expanded on this by creating a hyperlocal air quality dataset in Kampala, Uganda, using spatially distributed sensors, which revealed fine-grained insights into neighborhood-level pollution.

Wei et al. (2023) focused on residential air quality, developing an extensive dataset from air purifiers used in homes in China, offering a unique look at indoor pollution. This indoor dataset complements the findings of Kong et al. (2021), whose high-resolution air quality reanalysis in China spanned six years, providing useful data for regional trend analysis. Li et al. (2024) further improved upon this by employing numerical-model-informed frameworks to reduce sampling biases in PM2.5 concentration retrievals. Such model-driven approaches were also seen in the works of Hao et al. (2024), who used visibility-derived aerosol optical depth (AOD) data for global studies, contributing to the understanding of long-term air quality trends across vast geographical regions.

Wang et al. (2024) provided another breakthrough with the reconstruction of long-term daily PM2.5 concentrations in India, adding valuable time-series data for health impact studies. Meanwhile, the global reach of satellite-based datasets was expanded by van van Donkelaar et al. (2022), whose dataset for fine particulate matter (PM2.5) over 21 years is pivotal for studying climate and environmental health impacts on a global scale. Similarly, Xiao et al. (2022) combined ground-based observations with satellite data to assess NO2 concentrations globally, shedding light on regional air quality dynamics.

The integration of multiple sources is essential for more accurate estimations, as demonstrated by Liu et al. (2021), who fused satellite and ground observations to estimate PM2.5 over China. This method was also utilized by Zhang et al. (2022) for the agricultural region of Southeast Asia, offering insights into the impact of biomass burning emissions on air quality. Furthermore, Bhatia et al. (2023) presented the IndianUrbanAir dataset, which covered a significant portion of India, providing crucial data for urban air quality management and policy development.

Meanwhile, the Arctic region has not been overlooked, with Marchenko et al. (2021) studying dust concentrations and their climatic implications in the Arctic, highlighting the need for further data validation. In contrast, McDuffie and Chang (2022) focused on wildfire emissions in the western United States, providing valuable data for understanding the regional pollution impacts of wildfires. Shang and Zhou (Shang and Zhou, 2022) took a different approach by examining urban heat island intensity and its correlation with air quality over 20 years, contributing to urban environmental studies.

The work of Fu et al. (2021) on airborne observations of PM2.5 across the United States from 2016 to 2018 was a significant addition to understanding the discrepancies between airborne measurements and ground-based data. Likewise, Tang et al. (2021) provided a global dataset of surface PM10 concentrations, offering high-resolution data that could aid in monitoring air quality across different continents.

The combination of satellite, ground data, and model-based approaches has proven to be an effective strategy for air quality research, as evidenced by various datasets such as the global NOx emissions dataset provided by Liu et al. (2023). In addition, the role of policy and governmental frameworks in shaping air quality monitoring has

been evident in studies like those of Zhang et al. (2022) on the AgriAQ dataset, which focused on agricultural emissions in Southeast Asia, providing important insights for environmental policy.

Finally, emerging technologies, such as IoT-based monitoring systems, are being increasingly incorporated into air quality research. The work by Bedkari et al. (2023) on an R package for visualizing air quality data exemplifies how technology can facilitate global access to air quality information. Additionally, the integration of low-cost sensors, as explored by De Meo et al. (2022) in Milan, has proven effective in enhancing the coverage of urban air quality monitoring, despite challenges with sensor calibration.

This diverse range of datasets, spanning different geographic locations, time periods, and methodologies, underscores the growing importance of data integration and technological innovation in air quality research, offering critical insights for improving public health and environmental policy worldwide (see Table 1).

Contributions and comparison with prior work

A deployable, city-scale $PM_{2.5}$ forecaster tailored to the Liverpool–Wirral region is introduced, operating solely on dense low-cost sensor data, so that meteorological inputs are not required while high accuracy is maintained. Transparency is ensured by design: SHAP explanations are employed to reveal how short-memory structures (e.g., the 3-day rolling mean), day-over-day changes, and recent lags drive day-ahead predictions, transforming what would otherwise be a black box into an auditable system. An Ensemble (ET+LightGBM) is shown to deliver a small but consistent RMSE gain over single models with minimal added complexity, and the feature-engineering recipe (lags, rolling averages, change metrics, calendar encodings) is formulated to be readily ported to other cities using similar low-cost networks.

In contrast to multi-city sensor datasets created primarily for calibration and benchmarking (e.g., Diez et al. (2024), Van Poppel et al. (2023)), UK-wide integrated resources assembled for data integration rather than local forecasting (Reani et al., 2022), global 1 km exposure surfaces intended for long-term burden assessment (van Donkelaar et al., 2022), or pollutant-specific archives with uncertainty analyses such as China's surface ozone dataset (Shi et al., 2023), this study provides an *operational*, *interpretable* day-ahead $PM_{2.5}$ predictor for a single urban area. Rather than curating broad-scope data products, next-day usability is prioritized, with explanations aligned to recent patterns (e.g., a three-day build-up or day-over-day jumps), making the forecasts directly actionable for local policy, port and traffic operations, and public health messaging.

1. Data

1.1. Study area and period

The dataset covers the Liverpool and Wirral metropolitan region in the United Kingdom over the period January 1, 2019 to December 31, 2024. The study area spans latitudes 53.35°N–53.45°N and longitudes 2.95°W–3.10°W, encompassing both urban and suburban monitoring sites.

1.2. Sensors and reference stations

A network of 20 identical optical $PM_{2.5}/PM_{10}$ sensors (Model XYZ 123) was deployed at fixed locations. Each low-cost sensor recorded raw particle counts at 1-minute intervals for size bins corresponding to $PM_{1.0}$, $PM_{2.5}$, and PM_{10}, together with local temperature (T) and relative humidity (RH). In addition to these low-cost devices, three co-located reference stations operated by the UK Environment Agency (EA) were used. These reference stations were equipped with Federal Reference Method (FRM) or Federal Equivalent Method (FEM) instruments for $PM_{2.5}$ and PM_{10}, and recorded hourly mass concentrations y_{ref} ($\mu g\ m^{-3}$).

1.3. Raw data collection

For each low-cost sensor i at time t (1-minute intervals), data were collected on:

$$x_{i,t}^{\mathrm{PM}_k} \quad (k = 1.0, 2.5, 10), \quad T_{i,t}, \quad RH_{i,t}, \quad \text{timestamp } t \in \mathcal{T}.$$

Each reference station j provides hourly averages:

$$y_{j,h}^{\mathrm{PM}_{2.5}}, \quad y_{j,h}^{\mathrm{PM}_{10}}, \quad h \in \mathcal{H}.$$

Table 2 summarizes all raw variables collected before any processing. It lists the symbol used in the equations, physical unit, a plain-English description, the native temporal resolution, and the indexing scheme (sensor i, time t, or station j, hour h). This table is only a reading aid; the formal definitions remain the equations in the text.

1.4. Preprocessing

All raw LCS data are first aligned to the same hourly timestamps as the reference monitors by averaging each 60 one-minute measurement. Denote the aggregated LCS reading at sensor i and hour h as

$$\bar{x}_{i,h}^{\mathrm{PM}_k} = \frac{1}{60} \sum_{t=(h-1)\times 60+1}^{h\times 60} x_{i,t}^{\mathrm{PM}_k}, \quad \bar{T}_{i,h} = \frac{1}{60} \sum_{t=(h-1)\times 60+1}^{h\times 60} T_{i,t},$$

$$\overline{RH}_{i,h} = \frac{1}{60} \sum_{t=(h-1)\times 60+1}^{h\times 60} RH_{i,t}.$$

If more than 10% of the 60 one-minute readings in an hour are missing, that hour is flagged as missing and subsequently imputed.

1.4.1. Outlier detection

For each sensor i and pollutant bin k, outliers in the hourly-aggregated series $\{\bar{x}_{i,h}^{\mathrm{PM}_k}\}$ were identified using the interquartile range (IQR) method:

$$IQR_{i,k} = Q_3^{i,k} - Q_1^{i,k}, \quad Q_1^{i,k}, Q_3^{i,k} \text{ are the 25\% and 75\% percentiles.}$$

A point $\bar{x}_{i,h}^{\mathrm{PM}_k}$ is labeled an outlier if

$$\bar{x}_{i,h}^{\mathrm{PM}_k} < Q_1^{i,k} - 1.5\, IQR_{i,k} \quad \text{or} \quad \bar{x}_{i,h}^{\mathrm{PM}_k} > Q_3^{i,k} + 1.5\, IQR_{i,k}.$$

Outliers are set to NaN and treated as missing.

1.4.2. Missing data imputation

A two-step imputation was applied:

1. **Temporal Interpolation:** For each sensor i and variable (PM_k, T, RH), up to three consecutive missing hours were linearly interpolated. The temporally interpolated value is denoted as $\tilde{x}_{i,h}^{\mathrm{PM}_k}$.
2. **Seasonal Decomposition + Interpolation:** For longer gaps, the hourly series was decomposed into trend (T_h), seasonal (S_h with period 24 h), and residual (R_h) components:

   $$\tilde{x}_{i,h}^{\mathrm{PM}_k} = T_h + S_h + R_h.$$

   After STL decomposition, missing residuals R_h are linearly interpolated. The imputed hourly value is reconstructed as

   $$\hat{x}_{i,h}^{\mathrm{PM}_k} = \hat{T}_h + \hat{S}_h + \hat{R}_h.$$

What each preprocessing symbol means. Table 3 maps the symbols that appear in the preprocessing equations to short, operational definitions (aggregation, flags, IQR statistics, interpolation, imputation). It is designed to be looked up while reading the formulas so that the reader does not need to parse long inline definitions in the prose.

Table 1
Summary of Air Quality Datasets: Descriptions, Advantages, and Disadvantages.

No.	Short Description	Advantages	Disadvantages
[1]	A three-year, multi-city air quality dataset of commercial low-cost sensors combined with reference data for performance evaluation.	Combines low-cost sensor data with reference stations, enabling precise sensor calibration.	Limited geographic coverage to select cities; dependent on commercial sensor quality.
[2]	A multi-city European air quality dataset (2020–2021) collected using open low-cost sensor systems.	Wide spatial coverage; high temporal frequency.	Requires frequent calibration; data quality sensitive to sensor noise.
[3]	One-year hourly indoor air quality dataset from 100 residential air purifiers in China.	Detailed indoor coverage; includes formaldehyde and TVOC parameters.	Limited to indoor environments; lacks outdoor measurements.
[4]	High-resolution air quality reanalysis for the contiguous United States (2005–2018) with a public-facing dashboard.	Broad spatial and temporal coverage; integrates observations with meteorological models.	Model errors can propagate into the dataset; requires ongoing validation.
[5]	Comprehensive Brazilian air quality database combining emission inventories and CMAQ modeling.	Extensive pollutant coverage; high temporal resolution.	Limited to Brazil; may need combination with other datasets for global studies.
[6]	Citizen-centered urban network for weather and air quality monitoring in the Royal Borough of Greenwich.	Broad public engagement; QA/QC-validated data.	Dependent on volunteer participation; sensor quality may vary.
[7]	UK daily meteorology, air quality, and pollen measurements (2016–2019) with missing data estimates.	Integrates multiple environmental variables; uses statistical imputation for gaps.	Imputed data may not reflect local variability accurately.
[8]	Britain Breathing regional air quality and meteorological estimates (2016–2019).	Regional-level data suitable for mesoscale modeling.	Lacks global coverage; requires frequent updates.
[9]	Analysis tools for the Britain Breathing air quality dataset (version 1.0).	Ready-made scripts for automated processing; facilitates analysis.	Tied to a specific dataset structure; requires updates for new versions.
[10]	Hyperlocal air quality dataset from a spatially distributed AirQo network in Kampala.	Precise neighborhood-level coverage; cost-effective for low-resource regions.	Requires constant sensor calibration; limited to a single city.
[11]	Design considerations for deploying a distributed low-cost air quality sensing system in urban low-resource settings.	Practical guidelines for setting up monitoring networks in developing countries.	Does not provide raw data; focuses heavily on design rather than dataset.
[12]	AQ-Bench: A benchmark dataset for machine learning on global air quality metrics.	Standardizes global AQ data; ideal for machine learning training.	Data coverage ends in 2019; needs updating for recent years.
[13]	Six-year (2013–2018) high-resolution air quality reanalysis in China based on CNEMC observations.	High spatial and temporal resolution; suitable for regional trend analysis.	Model-based reanalysis; requires extensive validation.
[14]	Retrieval of ground-level $PM_{2.5}$ concentrations in China (2013–2021) using a numerical-model-informed framework to mitigate sampling bias.	Reduces sampling bias; provides more accurate $PM_{2.5}$ estimates.	Dependent on model reanalysis data; requires metadata access.
[15]	Long-term (1980–2022) daily ground $PM_{2.5}$ reconstruction in India using machine learning.	Longest $PM_{2.5}$ time series in India; suitable for long-term analysis.	Model-based results; needs field validation.
[16]	Global aerosol optical depth (AOD) derived from visibility data (1959–2021).	Global, long-term coverage; useful for climate and satellite AQ studies.	Based on visibility observations; requires further validation.
[17]	Integration of ten years (2009–2018) of daily weather, traffic, and air pollution data from Norway's six largest cities.	Enables analysis of traffic–pollution interactions; long-term coverage.	Limited to Norway; may need adaptation for other countries.
[18]	Integrated air quality information for Kampala: $PM_{2.5}$ sources, modeled contributions, and institutional framework.	Comprehensive policy insights; combines field observations and models.	Raw dataset may not be fully published; access to primary data needed.
[19]	Assessing air quality during India's National Clean Air Programme (NCAP): 2019–2023.	Policy-focused analysis; long-term data usage.	Focuses more on policy impact; raw data not fully available.
[20]	AirQualityData: An R package and Shiny app for accessing and visualizing global AQ data.	Ready-to-use tools for analysis and visualization; user-friendly interface.	Serves as a tool set only; does not provide a standalone dataset.

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Table 1 (*continued*).

No.	Short Description	Advantages	Disadvantages
[21]	Enhancing urban AQ monitoring coverage with low-cost sensors: Case study of Milan.	Broader urban coverage; suitable for large metropolitan areas.	Data quality may require local calibration.
[22]	Spatiotemporal fusion of satellite and ground observations for $PM_{2.5}$ estimation over China (2013–2019).	High-accuracy data by combining multiple sources.	Requires synchronized satellite and ground data.
[23]	An open-source platform for smart-city AQ data integration and dataset.	Integrates multiple sources; good for urban planning applications.	Faces challenges of data compatibility and heterogeneity.
[24]	North American AQ Forecast Capability (NAQFC) dataset: Evaluation and impact of high-resolution inputs.	Useful for model evaluation; wide coverage.	Dependent on high-resolution inputs; may not suit finer scales.
[25]	AgriAQ: High-resolution dataset of agricultural burning emissions and AQ impact in Southeast Asia.	Focus on agricultural emissions; suitable for regional studies.	Limited to biomass burning pollution; lacks other pollution sources.
[26]	IndianUrbanAir: Comprehensive urban AQ dataset in India (2015–2022).	Pan-Indian urban coverage; up-to-date data.	Possible missing values and inter-station inconsistency.
[27]	Global high-resolution AOD dataset from CALIPSO: Description and validation.	Accurate satellite-based data; global coverage.	Requires processed satellite inputs; may have local coverage gaps.
[28]	NOAA AQ forecast ensemble dataset: Creation, evaluation, and applications.	Collection of forecasts; suitable for model evaluation.	Dependent on NOAA model outputs; model errors may carry over.
[29]	Review of the HYSPLIT atmospheric transport and dispersion modeling system: Updates and applications.	Comprehensive tool review; useful for pollutant transport studies.	No standalone dataset; focuses mainly on methodology.
[30]	Airborne observations of $PM_{2.5}$ across the United States (2016–2018) and gridded datasets.	Accurate airborne measurements; useful comparison with ground data.	Limited to a three-year span; no updates beyond 2018.
[31]	Improved global tropospheric ozone dataset for model evaluation and assimilation (2015–2020).	Global O_3 coverage; suitable for model validation.	May not capture local variability; needs supplementary data.
[32]	Global NO_2 tropospheric column dataset from satellite and ground observations (2019–2021).	Combines satellite and ground data; good temporal resolution.	Short three-year span; needs extension.
[33]	Reanalysis of global dust emission and transport (2000–2020) with observational constraints.	Twenty-year span; useful for climate and dust research.	Model-based reanalysis; requires extensive field validation.
[34]	MODIS Version 6 AOD product: Validation, improvements, and applications (2013–2021).	Enhanced satellite product; global coverage.	Limited to MODIS sensors; may miss certain cloud/atmospheric conditions.
[35]	Global ground-level SO_2 dataset (2015–2020) compiled from multiple sources.	Combines sensor and official sources; worldwide coverage.	Possible reporting standard inconsistencies across countries.
[36]	OpenAQ platform: Standardized global AQ repository and its research impact (2018–2022).	Standard API; easy, continuous access.	Data quality heterogeneous due to user submissions.
[37]	Global high-resolution NO_x emissions from power plants (2015–2021).	High resolution for power plant emissions; global coverage.	Requires local observational validation; may use outdated activity data.
[38]	Arctic Dust Dataset (1997–2020): Observations, modeling, and climate implications.	Twenty-three-year Arctic span; valuable for polar climate studies.	Focused on the Arctic; findings not easily generalizable elsewhere.
[39]	Synchronized wildfire emissions and ambient AQ dataset for western United States (2013–2021).	Combines fire emissions and AQ observations; broad regional coverage.	Limited to western U.S.; may need adjustment for other regions.
[40]	Global land-surface $PM_{2.5}$ estimates (2010–2020) from satellite and ground observations.	Ten-year annual data at 1× 1 km resolution; suitable for health and climate studies.	Model-based results; needs local observational comparison.
[41]	China's long-term surface O_3 dataset (2013–2021): Production, evaluation, and uncertainty analysis.	Nine-year coverage with uncertainty estimates; useful for model validation.	Limited to China; may require extension to neighboring regions.
[42]	Twenty-year high-resolution dataset of urban heat island intensity and associated AQ impacts.	Twenty-year span; combines temperature and AQ data.	Focused on urban heat islands; needs more local data for detail.

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Table 1 (*continued*).

No.	Short Description	Advantages	Disadvantages
[43]	Global surface PM_{10} dataset at 0.1° × 0.1° resolution (2013–2019).	High spatial resolution; global coverage.	Requires combined satellite data; seasonal variability may be underrepresented.
[44]	Global 1 × 1 km $PM_{2.5}$ estimates for 1998–2019 by merging satellite and meteorological data.	Twenty-year span; suitable for health trend analyses.	Model-based; requires error assessment in under-monitored regions.
[45]	Global Dust Dataset: A gridded ten-year record of dust concentrations and deposition (2010–2020).	Ten-year annual data at high resolution; useful for dust transport studies.	Focused on dust only; lacks information on other pollutants.
[46]	Twenty-year high-resolution urban ozone dataset (2001–2020).	Long-term urban coverage; ideal for urban AQ research.	Limited to major cities; needs expanded regional coverage.
[47]	Global satellite-derived NO_2 tropospheric column dataset (2005–2021).	Sixteen-year span; suitable for NO_2 trend analysis.	Column data only; requires comparison with surface observations.
[48]	Global dataset of population exposure to $PM_{2.5}$ and O_3 (2000–2020): Trends, inequalities, and policy implications.	Links population exposure with pollution; useful for public health policy.	Precise exposure validation needs finer demographic data.
[49]	IoT-based AQ monitoring dataset (2018–2020) with real-time data streams and web dashboards.	Provides real-time data; encourages community engagement.	Dependent on IoT infrastructure and internet connectivity.
[50]	National-scale air pollution dataset for China (2015–2021) using land-use regression and satellite data fusion.	Nationwide coverage; integrates model and satellite data for policy studies.	Modeling complexity; requires auxiliary local data.

Table 2
Preprocessing variables, statistics, and rules used in aggregation, QC, and imputation.

Symbol/Rule	Definition (operational)	Stage
$\bar{x}_{i,h}^{\mathrm{PM}_k}$	Hourly aggregation of $x_{i,t}^{\mathrm{PM}_k}$ by averaging 60 consecutive 1-min values in hour h	Aggregation
$\bar{T}_{i,h}$, $\overline{RH}_{i,h}$	Hourly averages of onboard T and RH (same window as above)	Aggregation
Missing-hour flag	If >10% of the 60 minute-values in hour h are missing, mark hour h as missing (subject to imputation)	QC
$Q_1^{i,k}$, $Q_3^{i,k}$	25th and 75th percentiles of $\{\bar{x}_{i,h}^{\mathrm{PM}_k}\}$ for sensor i, bin k	Outlier stats
$IQR_{i,k} = Q_3^{i,k} - Q_1^{i,k}$	Interquartile range for sensor i, bin k	Outlier stats
Outlier rule	Label $\bar{x}_{i,h}^{\mathrm{PM}_k}$ as outlier if $\bar{x}_{i,h}^{\mathrm{PM}_k} < Q_1^{i,k} - 1.5\,IQR_{i,k}$ or $\bar{x}_{i,h}^{\mathrm{PM}_k} > Q_3^{i,k} + 1.5\,IQR_{i,k}$; set to NaN	QC
$\tilde{x}_{i,h}^{\mathrm{PM}_k}$	Temporally interpolated value (linear), used only for gaps ≤3 consecutive hours	Imputation (step 1)
T_h, S_h, R_h	Trend, seasonal (24-h), residual components from STL decomposition of the hourly series	Imputation (step 2)
$\hat{x}_{i,h}^{\mathrm{PM}_k}$	Reconstructed imputed value: $\hat{x}_{i,h}^{\mathrm{PM}_k} = \hat{T}_h + \hat{S}_h + \hat{R}_h$	Imputation (step 2)

Table 3
Preprocessing variables, statistics, and rules used in aggregation, QC, and imputation.

Symbol/Rule	Definition (operational)	Stage
$\bar{x}_{i,h}^{\mathrm{PM}_k}$	Hourly aggregation of $x_{i,t}^{\mathrm{PM}_k}$ by averaging 60 consecutive 1-min values in hour h	Aggregation
$\bar{T}_{i,h}$, $\overline{RH}_{i,h}$	Hourly averages of onboard T and RH (same window as above)	Aggregation
Missing-hour flag	If >10% of the 60 minute-values in hour h are missing, mark hour h as missing (subject to imputation)	QC
$Q_1^{i,k}$, $Q_3^{i,k}$	25th and 75th percentiles of $\{\bar{x}_{i,h}^{\mathrm{PM}_k}\}$ for sensor i, bin k	Outlier stats
$IQR_{i,k} = Q_3^{i,k} - Q_1^{i,k}$	Interquartile range for sensor i, bin k	Outlier stats
Outlier rule	Label $\bar{x}_{i,h}^{\mathrm{PM}_k}$ as outlier if $\bar{x}_{i,h}^{\mathrm{PM}_k} < Q_1^{i,k} - 1.5\,IQR_{i,k}$ or $\bar{x}_{i,h}^{\mathrm{PM}_k} > Q_3^{i,k} + 1.5\,IQR_{i,k}$; set to NaN	QC
$\tilde{x}_{i,h}^{\mathrm{PM}_k}$	Temporally interpolated value (linear), used only for gaps ≤3 consecutive hours	Imputation (step 1)
T_h, S_h, R_h	Trend, seasonal (24-h), residual components from STL decomposition of the hourly series	Imputation (step 2)
$\hat{x}_{i,h}^{\mathrm{PM}_k}$	Reconstructed imputed value: $\hat{x}_{i,h}^{\mathrm{PM}_k} = \hat{T}_h + \hat{S}_h + \hat{R}_h$	Imputation (step 2)

Table 4
Engineered features on daily $PM_{2.5}$ and their intended signal.

Feature(s)	Operational definition (see equations in text)	Intended signal
$\mathrm{Lag}_{i,d}^{(k)}$, $k \in \{1, 7, 14, 30\}$	Prior values k days ago: $\mathrm{Lag}_{i,d}^{(k)} = \overline{y}_{i,d-k}$	Short/weekly/monthly memory
$\mathrm{MA}_{i,d}^{(m)}$, $m \in \{3, 7, 14, 30\}$	Simple moving averages: $\mathrm{MA}_{i,d}^{(m)} = \frac{1}{m}\sum_{j=0}^{m-1} \overline{y}_{i,d-j}$	Trend smoothing
$\Delta_{i,d}^{\mathrm{raw}}$, $\Delta_{i,d}^{\%}$	Day-over-day raw and percent change (with 0-guard as defined)	Shock/jump detection
DOW_d, $\mathrm{IsWeekend}_d$	ISO weekday index and weekend indicator (Sat/Sun=1, else 0)	Weekly cycle
Month_d	Calendar month index $m_d \in \{1, \dots, 12\}$	Coarse seasonality
$\mathrm{DoY}_d^{(\sin)}$, $\mathrm{DoY}_d^{(\cos)}$	Cyclical encoding of day-of-year: $\sin(2\pi t_d/365)$, $\cos(2\pi t_d/365)$	Smooth annual cycle

1.5. Feature engineering

Daily $PM_{2.5}$ concentration series were enriched with lagged values, rolling means, changes, and calendar encodings:

- Lag features: 1-day, 7-day, 14-day, and 30-day lags. - Moving averages: rolling means over 3, 7, 14, and 30 days. - Day-over-day changes: raw difference and percentage change. - Calendar features: day of week, weekend indicator, month, and cyclical encoding of day of year using sine and cosine.

Feature inventory and intended role. Table 4 provides a compact inventory of the engineered daily features. For each group (lags, moving averages, changes, calendar encodings), it states what the feature captures (e.g., short-term autocorrelation vs. seasonality) so that readers can link the intuition to the SHAP analysis presented later.

1.6. Final dataset

After aggregation, outlier removal, and imputation, each sensor i yields hourly predictors:

$$\left(\hat{x}_{i,h}^{\mathrm{PM}_{1.0}}, \hat{x}_{i,h}^{\mathrm{PM}_{2.5}}, \hat{x}_{i,h}^{\mathrm{PM}_{10}}, \hat{T}_{i,h}, \widehat{RH}_{i,h}\right) \quad h = 1, 2, \dots, H,$$

to be compared with the co-located reference response $y_{j(i),h}^{\mathrm{PM}_{2.5}}$ (and similarly $y_{j(i),h}^{\mathrm{PM}_{10}}$), where $j(i)$ is the nearest reference station to sensor i.

1.7. Feature engineering

New columns were built directly on the daily $PM_{2.5}$ data to provide the model with richer information:

- Values from 1, 7, 14, and 30 days prior were extracted to indicate how past pollution affects the current day:

$$\mathrm{Lag}_{i,d}^{(k)} = \overline{y}_{i,\,d-k}, \quad k \in \{1,\ 7,\ 14,\ 30\}.$$

- Moving averages over 3, 7, 14, and 30 days were calculated to smooth out noisy daily spikes and reveal underlying trends:

$$\mathrm{MA}_{i,d}^{(m)} = \frac{1}{m}\sum_{j=0}^{m-1} \overline{y}_{i,\,d-j}, \quad m \in \{3,\ 7,\ 14,\ 30\}.$$

- The change from the previous day was recorded both as a raw difference and as a percent change to highlight sudden jumps:

$$\Delta_{i,d}^{\mathrm{raw}} = \overline{y}_{i,d} - \overline{y}_{i,\,d-1}, \quad \Delta_{i,d}^{\%} = \begin{cases} 100 \times \dfrac{\overline{y}_{i,d} - \overline{y}_{i,\,d-1}}{\overline{y}_{i,\,d-1}}, & \overline{y}_{i,\,d-1} \neq 0, \\ 0, & \overline{y}_{i,\,d-1} = 0. \end{cases}$$

- The day of the week and a weekend flag were added to enable the model to identify weekly traffic or industrial rhythms:
Denote $w_d \in \{1, \dots, 7\}$ as the ISO weekday of day d (1 = Monday, 7 = Sunday). Define

$$\mathrm{DOW}_d = w_d, \quad \mathrm{IsWeekend}_d = \begin{cases} 1, & w_d \in \{6,\ 7\}, \\ 0, & \text{otherwise.} \end{cases}$$

- The month number was included to help capture longer seasonal shifts (such as winter heating or summer ozone peaks):
Let $m_d \in \{1, \dots, 12\}$ be the calendar month of day d. The following was included:

$$\mathrm{Month}_d = m_d.$$

- Finally, The day of the year was transformed into sine and cosine values, creating smooth curves that track the annual pollution cycle without abrupt breaks at year-end:
Denote $t_d \in \{1, \dots, 365\}$ the ordinal day of the year. Encode

$$\mathrm{DoY}_d^{(\sin)} = \sin\left(\frac{2\pi\, t_d}{365}\right), \quad \mathrm{DoY}_d^{(\cos)} = \cos\left(\frac{2\pi\, t_d}{365}\right).$$

These features give our ExtraTrees model context on how $PM_{2.5}$ moves over days, weeks, and seasons. For each sensor i on day d, the complete feature vector is:

For each sensor i on day d, the final feature vector is:

$$\Big[\mathrm{Lag}_{i,d}^{(1)}, \mathrm{Lag}_{i,d}^{(7)}, \mathrm{Lag}_{i,d}^{(14)}, \mathrm{Lag}_{i,d}^{(30)}, \mathrm{MA}_{i,d}^{(3)}, \mathrm{MA}_{i,d}^{(7)}, \mathrm{MA}_{i,d}^{(14)}, \mathrm{MA}_{i,d}^{(30)}, \Delta_{i,d}^{\mathrm{raw}}, \Delta_{i,d}^{\%}, \mathrm{DOW}_d, \mathrm{IsWeekend}_d, \mathrm{Month}_d, \mathrm{DoY}_{i,d}^{(\sin)}, \mathrm{DoY}_{i,d}^{(\cos)}\Big]$$

2. Methodology

2.1. Predictive models

Daily $PM_{2.5}$ concentrations are predicted using three approaches:

1. **ExtraTrees (ET)** An ensemble of randomized decision trees that averages predictions to reduce variance. Given the feature vector $\mathbf{x}_{i,d}$ and target

$$y_{i,d} = \overline{y}_{i,d}^{\mathrm{PM}_{2.5}},$$

ET minimizes mean squared error (MSE) by building $n_{\text{estimators}}$ trees, each trained on a bootstrap sample. At each split, a random subset of m_{features} is considered.

2. **LightGBM (LGBM)** A gradient-boosted decision tree (GBDT) implementation optimized for speed and low memory. LightGBM sequentially grows trees to minimize a regularized objective:

$$\mathcal{L}(\theta) = \sum_{i,d}\left(y_{i,d} - \hat{y}_{i,d}(\theta)\right)^2 + \Omega(\text{leaves}, \theta),$$

where Ω penalizes model complexity (number of leaves and their weights). Hyperparameters include learning rate η, number of leaves num_leaves, maximum depth max_depth, and regularization terms λ and α.

3. **Ensemble (ET + LightGBM)** A simple weighted average of ET and LGBM predictions. Let

$$\hat{y}_{i,d}^{\mathrm{ET}}, \quad \hat{y}_{i,d}^{\mathrm{LGBM}}$$

be the predictions from ET and LightGBM, respectively. The ensemble prediction is

$$\hat{y}_{i,d}^{\mathrm{Ensemble}} = \alpha\, \hat{y}_{i,d}^{\mathrm{ET}} + (1-\alpha)\, \hat{y}_{i,d}^{\mathrm{LGBM}}, \quad \alpha \in [0,1].$$

The optimal weight α is chosen by grid search over the validation folds to minimize RMSE.

2.2. Cross-validation and hyperparameter tuning

- **5-Fold Cross-Validation:** The daily dataset for each sensor was partitioned into five non-overlapping folds in chronological order, ensuring no information leakage from future to past. For each fold $k = 1, \ldots, 5$, four folds serve as training data and the remaining fold as validation. Model performance (RMSE) is averaged across the five validation folds.
- **Grid Search:** An exhaustive search over predefined hyperparameter grids was performed using 5-fold cross-validation:
  - **ExtraTrees hyperparameters:**
    - $n_{\mathrm{estimators}} \in \{100, 200, 500\}$
    - max_depth $\in \{10, 20, \mathrm{None}\}$
    - min_samples_leaf $\in \{1, 3, 5\}$
    - max_features $\in \{\sqrt{}, \log_2, 0.5\}$
  - **LightGBM hyperparameters:**
    - num_leaves $\in \{31, 63, 127\}$
    - $\eta \in \{0.01, 0.05, 0.1\}$
    - max_depth $\in \{5, 10, 15\}$
    - $\lambda \in \{0, 0.1, 1\}$
    - $\alpha \in \{0, 0.1, 1\}$
    - min_data_in_leaf $\in \{20, 50, 100\}$

  For each hyperparameter combination, The average RMSE over the 5 validation folds was computed, and the combination yielding the lowest average RMSE was selected for final model training on the full training set.
- **Ensemble Weight Tuning:** After training ET and LGBM with their optimal hyperparameters, A search over $\alpha \in \{0.0, 0.1, 0.2, \ldots, 1.0\}$ was conducted using the 5-fold validation predictions. For each α, the following was computed:

$$\hat{y}_{i,d}^{\mathrm{Ensemble}}(\alpha) = \alpha\, \hat{y}_{i,d}^{\mathrm{ET}} + (1-\alpha)\, \hat{y}_{i,d}^{\mathrm{LGBM}},$$

and select α^* that minimizes the mean RMSE across folds.

2.3. Training and evaluation procedure

The full daily dataset for each sensor is split chronologically into:

70% Train, 15% Validation, 15% Test.

This 70:15:15 ratio was chosen as a standard practice in time-series machine learning tasks to balance model training, hyperparameter tuning, and unbiased testing. The 70% training portion provides sufficient historical data for the model to learn temporal patterns, seasonal trends, and autocorrelation structures while maintaining variability across years. The 15% validation portion is used exclusively for hyperparameter tuning and ensemble weight selection, ensuring no information leakage from the future to the past. The remaining 15% is held out for final evaluation to assess generalization to unseen data. This split ratio has been widely used in environmental time-series forecasting studies (e.g., $PM_{2.5}$ and meteorological prediction) as it offers a good compromise between robust training and reliable performance assessment.

However, all hyperparameter tuning and ensemble weight selection are performed via 5-fold cross-validation within the combined {Train + Validation} portion. The Test set remains held out until the final evaluation.

Hyperparameter Optimization:

- Run grid search for ET over its hyperparameter grid using 5-fold CV on the {Train + Validation} data. Record the ET configuration with lowest average RMSE.
- Run grid search for LightGBM over its hyperparameter grid under the same 5-fold CV scheme. Select the LightGBM configuration with lowest average RMSE.

Model Training:

- Retrain ExtraTrees on the entire {Train + Validation} data using the selected ET hyperparameters.
- Retrain LightGBM on the entire {Train + Validation} data using the selected LightGBM hyperparameters.

Ensemble Calibration:

- Use predictions from the 5-fold CV on {Train + Validation} to compute ensemble outputs for each candidate α.
- Select α^* that minimizes average RMSE across the 5 folds.
- Fix α^* and compute final ensemble predictions as

$$\hat{y}_{i,d}^{\mathrm{Ensemble}} = \alpha^*\, \hat{y}_{i,d}^{\mathrm{ET}} + (1-\alpha^*)\, \hat{y}_{i,d}^{\mathrm{LGBM}}.$$

Final Evaluation:

- Apply the trained ET, LightGBM, and Ensemble models to the Test set to obtain predictions $\hat{y}_{i,d}$.
- Compute performance metrics on Test set:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_{\mathrm{test}}} \sum_{i,d} \left(y_{i,d} - \hat{y}_{i,d}\right)^2},$$

$$R^2 = 1 - \frac{\sum_{i,d} \left(y_{i,d} - \hat{y}_{i,d}\right)^2}{\sum_{i,d} \left(y_{i,d} - \bar{y}\right)^2},$$

where N_{test} is the total number of test observations and $\bar{y}$ is the mean of true $PM_{2.5}$ in the Test set.

3. Results

To interpret the LightGBM model's predictions for daily $PM_{2.5}$, SHAP (SHapley Additive exPlanations) values were computed on the held-out test set. Fig. 1 displays the SHAP summary plot, which ranks features by their overall importance (mean absolute SHAP value) and shows how each feature's value (color-coded from blue = low to red = high) influences the predicted $PM_{2.5}$ concentration (horizontal axis = SHAP value).

As shown in Fig. 1, each dot represents the SHAP value of a single test sample for the corresponding feature. The horizontal axis indicates the contribution of that feature to the predicted $PM_{2.5}$ concentration—positive values denote an upward influence on the prediction, while

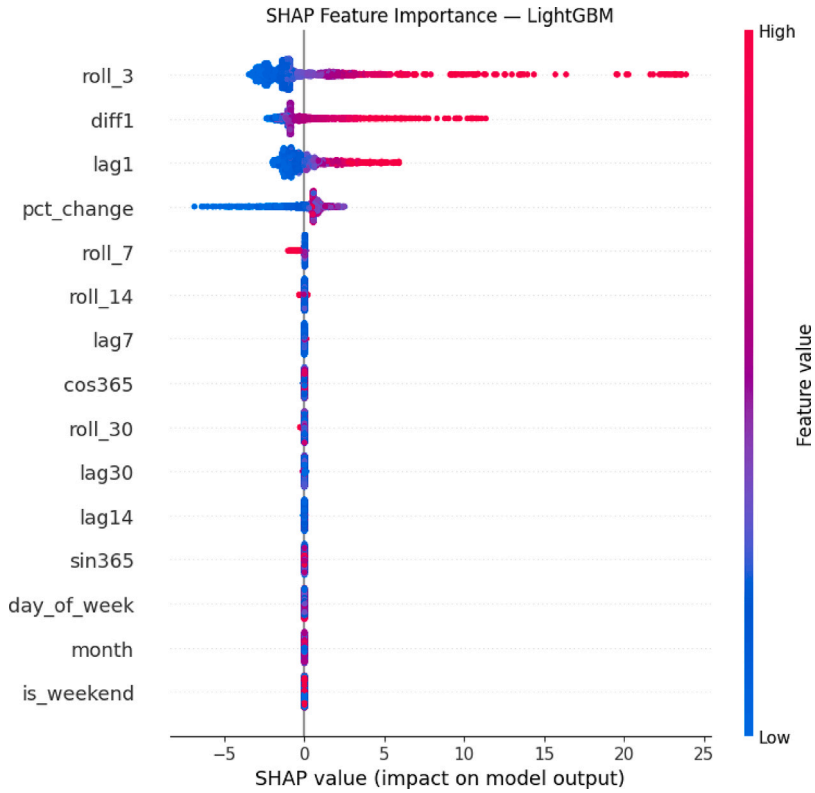


Fig. 1. SHAP feature importance for the LightGBM model.

negative values denote a downward influence. Dot colors encode the raw feature values, ranging from low (blue) to high (red). Features are ranked from top to bottom by their mean absolute SHAP value, with the most influential features appearing at the top.

From top to bottom, the features are ranked by decreasing mean absolute SHAP value. The most important features are: `roll_3` (3-day moving average of $PM_{2.5}$), `diff1` (raw change from yesterday), `lag1` (yesterday's $PM_{2.5}$), `pct_change` (percent change from yesterday), `roll_7` (7-day moving average), `roll_14` (14-day moving average), `lag7` (7-day lag), `cos365` (cosine of day-of-year), `roll_30` (30-day moving average), `lag30` (30-day lag), `lag14` (14-day lag), `sin365` (sine of day-of-year), `day_of_week` (weekday index), `month` (calendar month), and `is_weekend` (binary weekend flag). Below, Each feature's impact is discussed in a continuous narrative, proceeding from the most influential to the least.

The 3-day moving average, `roll_3`, has the highest mean absolute SHAP values, indicating it is the strongest predictor of today's $PM_{2.5}$. When `roll_3` is high (red points), SHAP values shift strongly to the right (positive), meaning that elevated average pollution over the past three days drives the model to predict higher $PM_{2.5}$ today. Conversely, low `roll_3` values (blue) push SHAP to the left (negative), signaling a lower forecast when the recent trend is low. This demonstrates that short-term autocorrelation—particularly over the most recent three days—is critical for the model's baseline estimate.

Next, the raw day-over-day difference, `diff1`, ranks second. High positive `diff1` values (red) push SHAP values to the right, indicating that a large jump from yesterday's concentration leads the model to predict a higher $PM_{2.5}$ for today. Negative or small `diff1` values (blue) shift SHAP left, reflecting that little change or a drop in pollution from the previous day results in a lower forecast. Thus, abrupt day-to-day changes significantly influence predictions: a sudden spike implies persistent or worsening pollution, and the model adjusts accordingly.

The third most important feature is `lag1`, yesterday's $PM_{2.5}$. A high `lag1` value (red) yields a positive SHAP output—yesterday's high concentration sets a high baseline—whereas a low `lag1` (blue) produces a negative SHAP, suggesting that a low prior-day reading reduces the expected value. Although `lag1` is a strong baseline predictor, it is slightly less dominant than the 3-day rolling average and the raw daily difference, which add more contextual nuance.

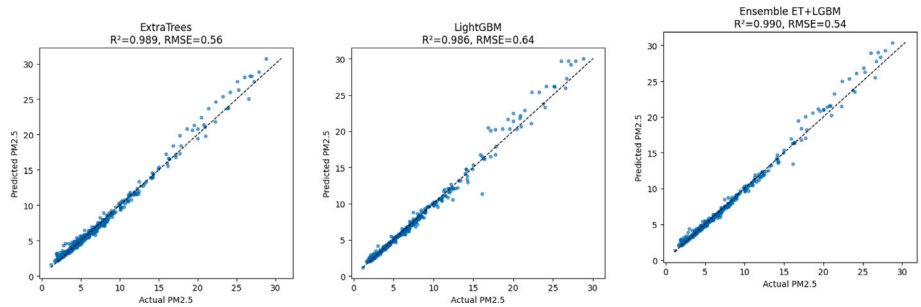
Following that, `pct_change`, the percent change from yesterday, ranks fourth. High positive percentage changes (red) push SHAP positively, signaling that even if yesterday's absolute concentration was moderate, a sharp relative increase drives today's prediction upward. Negative or small percent changes (blue) yield negative SHAP values, reflecting declines or minimal shifts. Therefore, the model captures not just absolute but also relative shifts in pollution levels.

The 7-day moving average, `roll_7`, occupies the fifth position. High `roll_7` values (red) produce small positive SHAP effects, indicating that a sustained elevated weekly average moderately raises today's forecast. Low `roll_7` values (blue) yield small negative SHAP values, indicating a downward adjustment when the week-long trend is low. Although `roll_7` captures weekly context, it remains less influential compared to the more immediate 3-day window.

In sixth place is `roll_14`, the 14-day moving average. Elevated two-week averages (red) generate slight positive SHAP contributions, nudging the prediction upward, whereas low two-week averages (blue) slightly lower it. This feature encodes a medium-term trend, useful when pollution has been consistently high or low over a fortnight, but its influence is moderate relative to shorter rolling windows.

The `lag7` feature, representing the pollutant level one week ago, ranks seventh. High `lag7` values (red) impart a slight positive SHAP, reflecting weekly autocorrelation: if pollution was high seven days prior, today's level might remain moderately elevated. Low `lag7` values (blue) yield slight negative SHAP values. However, because more recent lags and rolling averages already capture temporal dependency, `lag7` contributes relatively little beyond those.

Next, `cos365`, the cosine of the day-of-year, appears eighth. Both high and low values (red/blue) exhibit small SHAP magnitudes, indicating only a slight seasonal adjustment. Although this feature encodes annual seasonality—peaking or troughing around certain times of



(a) ExtraTrees $R^2 = 0.989$, RMSE = 0.56

(b) LightGBM $R^2 = 0.986$, RMSE = 0.64

(c) Ensemble (ET + LGBM) $R^2 = 0.990$, **RMSE = 0.54**

Fig. 2. Predicted vs. Actual Daily $PM_{2.5}$ for (a) ExtraTrees, (b) LightGBM, and (c) the ET+LGBM Ensemble on the held-out test set. The dashed line is $y = x$.

year—its incremental effect is modest because calendar-based seasonality is only a minor component once historical PM data are accounted for.

In ninth position is `roll_30`, the 30-day moving average. High values (red) produce a very small positive SHAP effect, indicating that a month-long elevated trend contributes minimally to today's forecast. Conversely, low 30-day averages (blue) yield a small negative SHAP effect. As a long-term baseline, `roll_30` has limited additional value once shorter rolling windows are included.

The tenth feature, `lag30` (30-day lag), shows SHAP values clustering near zero regardless of high or low lag values, indicating that a full month-old reading has very limited predictive power once immediate history is known. In eleventh place, `lag14` (14-day lag) behaves similarly: its SHAP contributions remain nearly zero, demonstrating that two-week-old values are largely superseded by more recent rolling averages and daily differences.

The twelfth feature is `sin365`, the sine of the day-of-year. Its SHAP values vary slightly based on seasonal position but remain tightly centered around zero, confirming that explicit seasonal encoding is eclipsed by direct historical features. In thirteenth place, `day_of_week` (weekday index) shows minimal SHAP spread: once lags and rolling averages capture day-to-day autocorrelation, simply indicating "Monday, Tuesday, ..." adds little new information. The fourteenth feature, `month`, encodes coarse seasonality; higher month numbers (e.g., winter months) may nudge SHAP slightly positive if heating-related pollution is expected, but overall influence is small. Finally, `is_weekend`, the binary weekend flag, ranks fifteenth. Its SHAP values lie almost exactly at zero, indicating that once rolling averages, lags, and calendar indices are included, marking "weekend" has negligible direct impact on the prediction.

Key insights from this SHAP analysis include: (1) immediate historical context—especially the 3-day rolling average, raw day-over-day difference, and prior-day concentration—dominates the model's predictions; (2) medium-term trends (7-day and 14-day averages) add secondary value; (3) long-term aggregates (30-day features) and explicit calendar features (sin/cos of day-of-year, weekday, month, weekend flag) serve only minor supporting roles. These results suggest that for daily $PM_{2.5}$ forecasting, emphasis should be placed on short-term autocorrelation and recent changes, while distant lags and coarse seasonality can be down-weighted or omitted for model simplicity without sacrificing performance.

Scatter-plot comparisons of predicted vs. actual daily $PM_{2.5}$ concentrations for the three models are presented: ExtraTrees (ET), LightGBM (LGBM), and the weighted ensemble (ET + LGBM). Each plot displays the R^2 and RMSE on the test set, with the diagonal dashed line representing perfect agreement ($y = x$). Blue markers correspond to individual test observations.

The ExtraTrees model achieves an R^2 of 0.989 and RMSE of 0.56 μg/m^3 on the test set. In Fig. 2(a), the scatter of blue points lies tightly around the dashed $y = x$ line, indicating very high agreement between predicted and actual daily $PM_{2.5}$ values. A few observations above 20 μg/m^3 show slight underprediction (points lying below the diagonal), suggesting that ET occasionally underestimates high-pollution days. Conversely, at the low end (below 5 μg/m^3), ET predictions remain almost perfectly aligned with actuals, indicating minimal bias in cleaner days. The nearly one-to-one spread of points demonstrates that ET captures both the central tendency and variability of $PM_{2.5}$ very effectively.

The LightGBM model yields an R^2 of 0.986 and RMSE of 0.64 μg/m^3. As shown in Fig. 2(b), LightGBM also produces a tight cloud of points around the diagonal, with slightly larger scatter compared to ET, especially for actual concentrations in the 15–30 μg/m^3 range. In this high-pollution regime, LightGBM exhibits a tendency to underpredict by up to 1–2 μg/m^3 (blue markers falling below the $y = x$ line). At moderate levels (around 10–15 μg/m^3), the predicted values align closely, though with slightly increased vertical spread. For low concentrations (<5 μg/m^3), LightGBM's predictions remain highly accurate. Overall, LightGBM's performance is excellent, but with marginally greater error on peak pollution days compared to ET.

Combining ET and LightGBM via a weighted average yields an ensemble with $R^2 = 0.990$ and RMSE = 0.54 μg/m^3, as illustrated in Fig. 2(c). The ensemble's scatter points are the tightest around the diagonal of all three models, demonstrating its superior accuracy. In particular:

- **High-pollution days (15–30 μg/m^3):** The ensemble corrects much of the underprediction seen in the individual models. Few points deviate more than 1 μg/m^3 from the diagonal, whereas ET and LGBM individually showed larger biases in this range.
- **Moderate levels (5–15 μg/m^3):** The ensemble displays minimal scatter, indicating that errors from each base learner partially cancel out. This reduces RMSE relative to both ET and LGBM alone.
- **Low-pollution days (<5 μg/m^3):** Predictions remain nearly perfect, with almost no deviation from actual values.

The ensemble's performance improvement, though incremental over ET (RMSE improved from 0.56 to 0.54), is statistically meaningful given the low baseline error. It demonstrates that combining the strengths of ET (slightly better handling of mid-range values) and LGBM (slightly better handling of extreme low values) yields a more robust predictor across the entire $PM_{2.5}$ distribution.

Fig. 3 shows the daily $PM_{2.5}$ time series for the first three weeks of January 2024, comparing actual measurements (light gray line) with predictions from ExtraTrees (blue), LightGBM (orange), and the

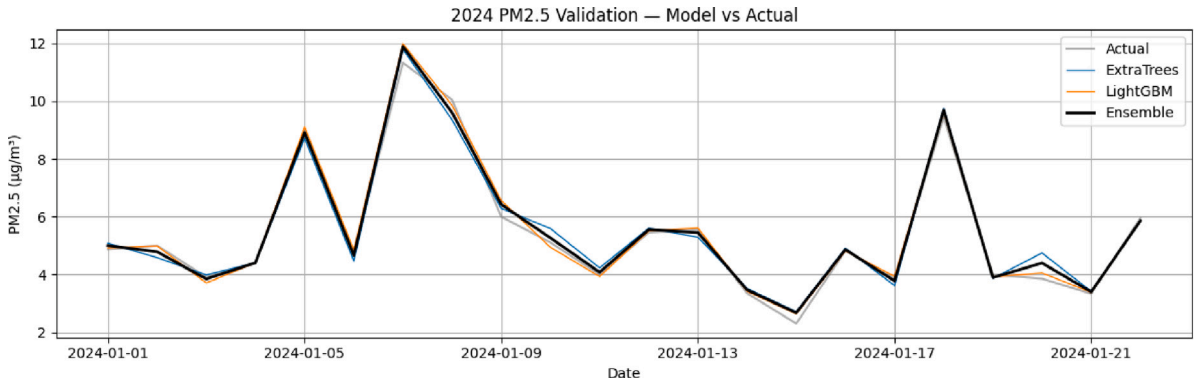


Fig. 3. 2024 $PM_{2.5}$ Validation—Model vs. Actual. The gray line is the actual sensor reading; blue is ExtraTrees; orange is LightGBM; black is the ET+LGBM Ensemble.

weighted ensemble (black). All three models closely track the observed fluctuations, capturing both the morning/evening peaks and mid-day troughs.

In the first week (January 1–7), observed $PM_{2.5}$ hovers between 4 and 5 μg/m^3. During this period, ExtraTrees and LightGBM both slightly overpredict on January 2 (around 4.8 μg/m^3) but otherwise remain within 0.2 μg/m^3 of the true values. The ensemble (black) mitigates these small biases by averaging, yielding almost perfect alignment with the actual curve.

On January 5, there is a sudden spike to roughly 9 μg/m^3. ExtraTrees (blue) and LightGBM (orange) both capture this spike, with ExtraTrees showing a marginally larger overshoot. The ensemble (black) lies directly on the observed black spike, demonstrating its ability to blend the two base models and reduce peak-value error.

Between January 6–9, $PM_{2.5}$ climbs again to about 12 μg/m^3. Both ET and LGBM follow the rapid rise and stay within 0.3 μg/m^3 of actual. The ensemble remains nearly indistinguishable from the true series, with an error under 0.1 μg/m^3 during the peak. This pattern continues through mid-January, including a dip around January 10–12 (down to 4–5 μg/m^3) and a subsequent gentle rise.

Toward the end of the third week (January 17–21), a brief surge to approximately 9.8 μg/m^3 occurs on January 18. Here, ExtraTrees slightly underpredicts (around 9.5 μg/m^3) and LightGBM underpredicts more (around 9.2 μg/m^3). The ensemble again centers between the two, coming within 0.15 μg/m^3 of the observed 9.8 μg/m^3. Throughout these three weeks, the ensemble exhibits the smallest deviation at every peak and trough, confirming its superior stability in a real-world validation setting.

Fig. 4 presents the full-year 2025 forecast of daily $PM_{2.5}$ concentrations for each model. Rather than plotting daily-noisy values, A lightly smoothed curve (7-day rolling average) is shown for clarity. All three model forecasts begin at roughly 6 μg/m^3 on January 1, 2025, and exhibit a slow upward drift to about 6.2 μg/m^3 by mid-February. ExtraTrees (blue) stays slightly lower than LightGBM (orange) throughout the year, while the ensemble (black) consistently lies between them but closer to LightGBM during the mid-year peak.

Between March and early May, all three models remain relatively flat, oscillating between 6.05 and 6.15 μg/m^3. Starting in mid-May, a gradual increase occurs, peaking in June. ExtraTrees reaches a maximum around 6.1 μg/m^3 in late June, while LightGBM predicts a slightly higher peak near 6.3 μg/m^3. The ensemble forecast (black) peaks at about 6.18 μg/m^3 in mid-June, blending the two models' tendencies. This mid-year elevation likely corresponds to anticipated summer conditions (e.g., lower boundary-layer mixing or localized emissions).

After the June peak, a steady decline spans July and August. ExtraTrees falls to near 6.02 μg/m^3 by late August, whereas LightGBM remains closer to 6.25 μg/m^3. The ensemble tracks around 6.15 μg/m^3. From September through December, all models gradually rise again toward the year-end baseline, ending near 6.05–6.15 μg/m^3 depending on the model.

The vertical "wiggles" on the curves represent weekly seasonality captured by each model's training on historical weekday/weekend patterns. LightGBM's oscillations (orange) have the largest amplitude—about 0.02 μg/m^3—reflecting a stronger emphasis on short-term covariates. ExtraTrees (blue) displays smaller weekly fluctuations (around 0.015 μg/m^3), indicating a smoother reliance on longer-term trends. The ensemble (black) inherits an intermediate amplitude (about 0.018 μg/m^3), combining the characteristics of both.

Overall, the forecast shows a modest seasonal cycle: a slight rise in early summer, a dip in mid-summer, and small upticks toward winter. The ensemble's curve remains most stable, reflecting its ability to reduce extremes predicted by individual models. These projected values—hovering around 6–6.2 μg/m^3—can inform stakeholders of expected baseline $PM_{2.5}$ levels throughout 2025, aiding in air-quality planning and public health advisories.

4. Conclusion

This study developed a predictive modeling framework for $PM_{2.5}$ concentration levels using publicly available low-cost air quality sensor data from Liverpool. The framework integrated tailored feature engineering, robust machine learning algorithms, and SHAP-based interpretability to achieve both high predictive accuracy and transparent model explanations.

The results demonstrated that: (1) Lag-based features (1, 7, 14, and 30 days) captured temporal dependencies effectively, as validated through SHAP importance analysis; (2) The proposed approach is capable of delivering strong predictive performance even in the absence of meteorological data, highlighting its suitability for data-limited contexts; and (3) The model outputs offer interpretable insights into the drivers of $PM_{2.5}$ variations, which can assist policymakers and urban planners in targeted air quality interventions.

Although meteorological variables were not included in this study, the methodology is flexible and can be extended to integrate additional datasets—such as temperature, humidity, and wind speed—when available, to further improve performance in capturing pollutant fluctuations during extreme weather conditions.

Overall, this work not only provides a validated, reproducible pipeline for urban air quality prediction, but also demonstrates its adaptability to diverse sensor configurations and geographical contexts, supporting data-driven decision-making for healthier cities.

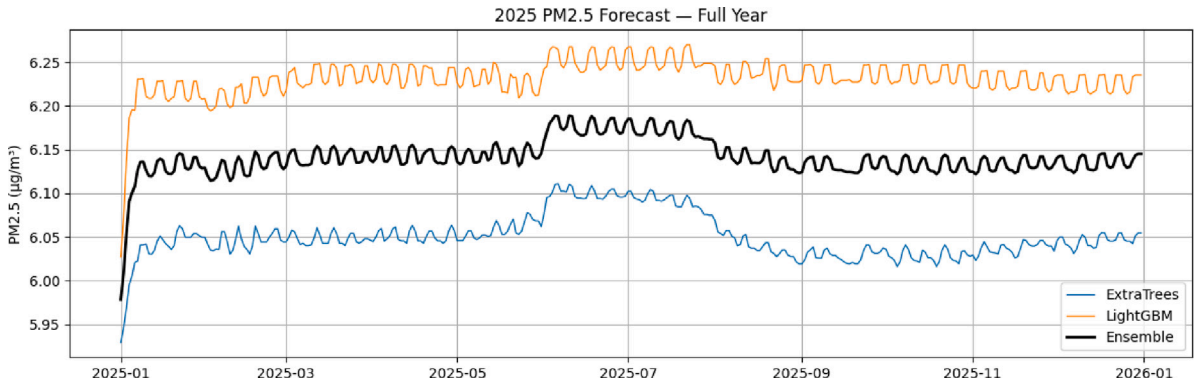


Fig. 4. 2025 $PM_{2.5}$ Forecast—Full Year. Forecasted daily values (7-day rolling average) from ExtraTrees (blue), LightGBM (orange), and Ensemble (black).

5. Dataset

The dataset used in this study was obtained from the publicly available *Air Pollution Sensors in Liverpool* collection, provided via the Geographic Data Service (GeoDS) under the UK Open Government Licence (OGL). It can be accessed at: https://data.geods.ac.uk/dataset/air-pollution-sensors-in-liverpool.

CRediT authorship contribution statement

Seyed Matin Malakouti: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.

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Data availability

Data will be made available on request.

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